

# 1. HOSTS, CLIENTS, SERVERS AND PEERS

## Roles without memorizing boxes



**GOAL:** Identify the role each host plays in one exchange and test the correct endpoint.

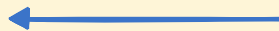
### SAME HOSTS, DIFFERENT ROLES PER EXCHANGE

#### 1 WEB BROWSING: CLIENT → SERVER



Laptop  
**CLIENT**  
Web browser  
(e.g. Chrome)

Request:  
GET /index.html



Response:  
web page (HTML)

Web server  
**SERVER**  
Web service  
(e.g. nginx, Apache)

#### 2 PHONE BACKUP: CLIENT → SERVER



Phone  
**CLIENT**  
Backup app  
(e.g. iCloud, Google Drive)

Request:  
upload photos



Response:  
upload successful

Laptop  
**SERVER**  
Backup service  
(e.g. file server, sync app)

#### 3 PEER-TO-PEER: BOTH CLIENT AND SERVER



Laptop A  
**PEER**  
File sharing app  
(e.g. Syncthing)

Request: ask for a file block

Response: send the file block

... and later the other way ...



Laptop B  
**PEER**  
File sharing app  
(e.g. Syncthing)

### QUICK DEFINITIONS

#### HOST

- A network-connected system with one or more interfaces (network card, Wi-Fi, virtual).

#### CLIENT

- Starts a request for a service.

#### SERVER

- Listens and responds with a service.

#### PEER

- An endpoint that acts as both client and server in the same system.

### A BOX IS NOT A ROLE



Web browser  
(client)



File server  
(server)

One physical host can run client and server processes at the same time. Roles depend on the exchange, not on the device type.

### WORKED EXAMPLE: FOLLOW ONE EXCHANGE

#### 1. USER ACTION

You open  
`https://example.com`  
in your browser.



#### 2. CLIENT PROCESS

Web browser  
(client), e.g. Chrome



#### 3. ADDRESS + PORT

DNS gives the  
server IP; HTTPS  
uses port 443.



#### 4. SERVER PROCESS

Web server  
(server), e.g. nginx



#### 5. RESPONSE

Web page data (HTML,  
images, CSS, etc.)



### DIAGNOSTIC HABIT:

Name the process, direction and requested service before blaming a device.

- Which process made the request?
- Is it going to a client, server or peer?
- Which service and port is used?
- What response do you expect?



### REMEMBER:

Client and server describe behaviour in an exchange, not permanent machine types.